A type-II_∞ modular-shadow algebra for the Bianchi IX vacuum sector via Manin–Marcolli symbolic dynamics

(conjecture paper, with falsifier protocol)

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Abstract

We propose a type-II_∞ modular-shadow algebra $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ for the vacuum sector of Bianchi IX cosmology in the Belinski–Khalatnikov–Lifshitz (BKL) mixmaster phase, obtained as the Speranza (2025) crossed product of a wedge-complement type-III_1 field algebra $\mathcal{N}$ by the Connes–Takesaki modular automorphism flow. Combining Manin–Marcolli's (2015) symbolic-dynamics representation of the BKL Kasner bounce sequence as the geodesic flow of the modular surface $X(2) = \Gamma(2)\backslash\mathbb{H}$ with the generic crossed-product framework of Witten (2022) and Speranza (2025), we *conjecture* that the modular flow rate on $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ equals $\lambda_{\mathrm{mod}} = 2\pi\,\lambda_{\mathrm{BKL}}$, where $\lambda_{\mathrm{BKL}} = h_{\mathrm{KS}}(\sigma_{\mathrm{Gauss}})$ is the Kolmogorov–Sinai entropy of the Gauss-shift symbolic encoding of BKL Kasner bounces. This in turn predicts a Maldacena–Shenker–Stanford-style modular Krylov-complexity bound $\mathcal{C}_K^{\mathrm{mod}}(s) \le A\,\cosh(2\pi\,\lambda_{\mathrm{BKL}}\,s)$ on $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$. The construction is presented as four explicit conjectures (Conjectures M45.1.A–M45.1.D) and a one-sided numerical falsifier based on direct computation of λ_{BKL} via the Gauss shift and comparison with Krylov-rate growth in a discretised Bianchi IX wave-function evolution. We do *not* prove any of the four conjectures; six explicit "[TBD: prove]" markers are flagged. The proposal is compatible with the Heinzle–Uggla (2009) Bianchi IX attractor theorem and with our companion work on the Modular Shadow MSS bound for the isotropic FLRW sector.

1 Introduction

The Maldacena–Shenker–Stanford (MSS) bound on chaos [1] states that the Lyapunov rate of any large-N thermal quantum system satisfies $\lambda_L \le 2\pi/\beta$. Its modular-Krylov refinement of Caputa–Magán–Patramanis–Tonni [3] and the universal operator-growth inequality of Parker–Cao–Avdoshkin–Scaffidi–Altman [2] give a sharper, large-N-free statement on the operator-Hilbert space of modular evolution.

In parallel, the program initiated by Witten [4, 5] and Chandrasekaran–Longo–Penington–Witten [6] has established that gravitational observable algebras with quantum reference frames are type-II_∞ crossed products $\mathcal{A}_{\mathrm{II}_\infty} = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$, restoring a semifinite trace.

Speranza [8] extended this to a generic *intrinsic* cosmological observer construction, applicable to backgrounds without a global timelike Killing vector. Kudler-Flam–Leutheusser–Satishchandran [7] concurrently worked out the FLRW (isotropic) case in detail.

The Bianchi IX vacuum, with its anisotropic SO(3)-symmetric metric and three independent scale factors, is the canonical *anisotropic* analog of FLRW. In the BKL approach to its initial-singularity behaviour, generic solutions exhibit a chaotic infinite sequence of Kasner epochs separated by bounces governed by a hyperbolic billiard in Misner variables [11, 12]. Manin–Marcolli [9] proved a remarkable correspondence:

$$\text{Mixmaster BKL bounce sequence} \quad \longleftrightarrow \quad \text{geodesic flow on } X(2) = \Gamma(2)\backslash\mathbb{H}, \tag{1}$$

realised by a two-sided continued-fraction shift, with Gauss measure as the unique invariant probability of maximal entropy. Fan–Fathizadeh–Marcolli [10] further showed that the Connes spectral action on Bianchi IX gravitational instantons admits modular-form expansions of weights 2, 14, 18.

De Clerck–Hartnoll–Santos [**?**] studied AdS Mixmaster chaos via the $\Gamma(2)$ arithmetic substrate (arXiv:2312.11622), focusing on Hamiltonian-quantization eigenvalues of the interior geometry. Our framework provides the natural type-II_∞ crossed-product + Krylov-complexity completion of that programme: we identify the Speranza modular flow with the BKL Gauss-shift rate, and conjecture a sharp $\lambda_K \leq \lambda_{\mathrm{BKL}}$ bound.

Aim of this paper. We propose a *conjectural* synthesis of the two pictures: combine the Speranza/CLPW crossed-product construction with the Manin–Marcolli geodesic-flow representation of BKL chaos, identify the resulting type-II_∞ algebra $\mathcal{M}^{\mathrm{II}_\infty}_{\mathrm{BIX}}$ as the *modular-shadow algebra* of the Bianchi IX vacuum sector, and *conjecture* that its modular flow rate equals 2π times the Kolmogorov–Sinai entropy λ_{BKL} of the Gauss shift. This in turn predicts an MSS-style Krylov-complexity bound at $\beta_{\mathrm{eff}} = 2\pi/\lambda_{\mathrm{BKL}}$.

What this paper does *not* prove. None of the four sub-conjectures (Conjectures 2–6) is proved. Six explicit "[TBD: prove]" markers are flagged in the construction sketch (§4). This is a *programmatic* note: it identifies a precise, testable target, sketches the construction, and presents a one-sided numerical falsifier (§7).

Honesty bar. The Manin–Marcolli correspondence (1) is a theorem; the type-II_∞ crossed-product framework of Witten/Speranza/CLPW is established; the MSS bound and its Krylov refinement are established. The *novelty* here lies entirely in identifying the rate of the Speranza modular flow with $2\pi\lambda_{\mathrm{BKL}}$. That identification is open.

Organization. §2 reviews the three preliminary frameworks. §3 states the central Conjecture M45.1. §4 sketches the four-step construction. §5 formulates the MSS-style Krylov bound. §6 relates to our Bianchi IX Lemma A.1 (paper #5). §7 presents a numerical falsifier protocol. §8 lists open problems.

2 Preliminaries: three frameworks

2.1 Manin–Marcolli BKL ↔ X(2) geodesic flow

Misner variables for Bianchi IX vacuum reduce the dynamics to the geodesic flow of an indefinite metric on a hyperbolic billiard table in $\mathbb{H}^2$ [11]. Cornish–Levin [12] demonstrated,

via observer-independent fractal methods, the existence of a strange repellor in the mixmaster dynamics. Manin–Marcolli [9] showed that the BKL Kasner-bounce sequence is conjugate to the two-sided shift $\sigma : \Sigma_2 \to \Sigma_2$ associated with the continued-fraction expansion, equivalently the geodesic flow on the modular surface $X(2) = \Gamma(2)\backslash\mathbb{H}$ along its imaginary axis. The unique invariant probability of maximal entropy for σ is the Gauss measure

$$d\mu_{\text{Gauss}}(x) = \frac{1}{\log 2}\,\frac{dx}{1+x}, \qquad x \in [0,1], \tag{2}$$

with topological entropy $h_{\text{top}}(\sigma) = \pi^2/6$ (the "Lochs–Khinchin" constant; see [9] §2.4).

Definition 1 (BKL Lyapunov rate)**.** *We denote by*

$$\lambda_{\text{BKL}} \;:=\; h_{\text{KS}}(\sigma, \mu_{\text{Gauss}}) \;=\; \frac{\pi^2}{6\log 2} \;\approx\; 2.3731, \tag{3}$$

the Kolmogorov–Sinai entropy of the Gauss shift with respect to μ_{Gauss} (Manin–Marcolli [9], Theorem 2.4 and [10]).

2.2 Witten II_∞ crossed product

Let $\mathcal{N}$ be the type-III_1 von Neumann algebra of gauge-invariant local field operators on a wedge-complement region of some background spacetime, in the cyclic-and-separating GNS representation of a KMS state ϕ at inverse temperature β. Tomita–Takesaki gives the modular automorphism group $\sigma_t^\phi : \mathcal{N} \to \mathcal{N}$. Witten [5] and CLPW [6] showed that the crossed product $\mathcal{N} \rtimes_{\sigma^\phi} \mathbb{R}$ is type-II_∞, semifinite, with trace dual to the modular flow. The Connes–Takesaki flow of weights identifies the natural rate of σ_t^ϕ with $2\pi/\beta$ (in normalised units).

2.3 Speranza intrinsic observer

For backgrounds without a global timelike Killing vector (de Sitter static patch, FLRW, Bianchi), Speranza [8] constructs an *intrinsic cosmological observer*: a worldline γ equipped with a quantum reference clock, and an associated crossed-product algebra

$$\mathcal{A}_\gamma^{\text{II}_\infty} \;=\; \mathcal{A}_{\gamma^c}^{\text{III}_1} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}, \tag{4}$$

where $\mathcal{A}_{\gamma^c}^{\text{III}_1}$ is the type-III_1 algebra of fields on the wedge complement of γ, and ϕ_γ is the geometric (vacuum-like) state for which a local modular automorphism is well-defined. Kudler-Flam–Leutheusser–Satishchandran [7] worked this out explicitly for FLRW backgrounds. For Bianchi IX, the construction has not yet been carried out; this gap is precisely what Conjectures M45.1.B and M45.1.C below address.

3 Central conjecture

Let M_{BIX} denote a Bianchi IX vacuum spacetime in mixmaster (BKL) phase, γ an intrinsic Speranza observer with worldline parametrised by Misner volume time τ, and $\mathcal{A}_{\text{obs}} := \mathcal{A}_{\gamma^c}^{\text{III}_1}$ the type-III_1 algebra of bounded gauge-invariant local field observables on the wedge complement of γ in M_{BIX}.

Conjecture 1 (Bianchi IX modular-shadow algebra, Conjecture M45.1)**.** *There exists a faithful normal representation $\pi : \mathcal{A}_{\rm obs} \to B(\mathcal{H}_{\rm BIX})$ on a Hilbert space $\mathcal{H}_{\rm BIX}$ unitarily equivalent to $L^2(X(2), \mu_{\rm Gauss}) \otimes \ell^2$, such that:*

(i) the Speranza crossed product $\mathcal{M}^{\rm II_\infty}_{\rm BIX} := \mathcal{A}_{\rm obs} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}$ is a type-II_∞ factor with semifinite trace $\tau_{\rm II}$;

(ii) the modular automorphism $\sigma_t^{\phi_\gamma}$ is implemented (up to inner automorphism) by the Manin–Marcolli geodesic flow g_t on $L^2(X(2), \mu_{\rm Gauss})$;

(iii) consequently, the natural modular flow rate is $\lambda_{\rm mod} = 2\pi\, \lambda_{\rm BKL}$ with $\lambda_{\rm BKL}$ from Definition 1.

We split this into four sub-conjectures (§4) and derive the falsifiable Krylov bound (Conjecture 6, §5).

4 Sketch of construction

4.1 Step A: Wedge algebra setup

Setup

Let $M_{\rm BIX} = \mathbb{R} \times S^3$ carry a Bianchi IX vacuum metric

$$ds^2 = -N^2(\tau)\, d\tau^2 + \tfrac{1}{4}\, e^{-2\Omega(\tau)} \left(e^{2\beta(\tau)}\right)_{ij} \sigma^i \otimes \sigma^j, \tag{5}$$

where τ is Misner volume time ($\Omega = -\frac{1}{3} \ln V$, monotone in the BKL backward direction), $\{\sigma^i\}$ are the left-invariant 1-forms on $S^3 \cong \mathrm{SU}(2)$, and $\beta = \beta_+ \,\mathrm{diag}(1, 1, -2) + \beta_- \,\mathrm{diag}(1, -1, 0)$ is the Misner anisotropy matrix. The spatial sections are homogeneous under the SO(3) left-action. In the BKL (Mixmaster) phase approaching the singularity $\tau \to +\infty$, the metric (**??**) undergoes an infinite sequence of Kasner epochs separated by bounces encoded by the Gauss-shift map $\sigma_G(x) = \{1/x\}$ [9].

Fix an intrinsic observer $\gamma : \mathbb{R} \to M_{\rm BIX}$ with worldline parametrised by τ. Define the *wedge complement*

$$W_\gamma = \{\, p \in M_{\rm BIX} \;:\; p \text{ is spacelike-separated from every point on } \gamma \,\} \tag{6}$$

(well-defined locally, since the homogeneous SO(3) foliation provides a canonical spacelike splitting [13]). Consider a free real scalar field ϕ on $M_{\rm BIX}$ satisfying the covariant Klein–Gordon equation $\Box_g \phi = m^2 \phi$; the graviton case is analogous but algebraically more involved. Let $\mathcal{A}(W_\gamma)$ be the von Neumann algebra of bounded operators generated by the Weyl operators $W(f) = e^{i\phi(f)}$ with $\mathrm{supp}(f) \subset W_\gamma$, in the GNS representation of a reference state $\omega_{\rm BIX}$ (specified below).

Conjecture 2 (Wedge algebra type, M45.1.A — TBD-1)**.** *$\mathcal{A}(W_\gamma)$ is the unique hyperfinite type-III_1 factor $\mathcal{R}_\infty$.*

Why the standard arguments do not apply. The classical route to type-III_1 for wedge algebras runs through the *Bisognano–Wichmann theorem* [**?**]: in Minkowski space, the modular Hamiltonian of the Rindler wedge in the Minkowski vacuum is the *boost generator* K, a global Killing vector field. By Tomita–Takesaki theory, this forces the modular automorphism group $\sigma_t^\omega(A) = e^{itK}Ae^{-itK}$ to be a continuous one-parameter group of outer automorphisms, and the associated Connes spectrum $S(\mathcal{A}) = [0, \infty)$, which characterises type-III_1 [5]. For the de Sitter static patch, Chandrasekaran–Longo–Penington–Witten [6] show the algebra is the unique hyperfinite III_1 factor $\mathcal{R}_\infty$, again by identifying the modular Hamiltonian with the de Sitter isometry generator (a Killing vector of the static patch).

Neither argument transfers to Bianchi IX. The metric (**??**) admits *no global timelike Killing vector*: the three spatial Killing vectors $\{\xi_i\}$ generating the SO(3) symmetry are spacelike, and the Misner volume τ is not a Killing coordinate (it appears only as a synchronous gauge choice). Consequently, the modular Hamiltonian of $\mathcal{A}(W_\gamma)$ is *not* identified with any known geometric symmetry generator, and the Bisognano–Wichmann route is unavailable.

We note also that the FLRW case handled by Kudler-Flam–Leutheusser–Satishchandran [7] follows a *different* path: the FLRW algebra is shown to be type-II_∞ *directly* (via the inflaton zero-mode and the KMS condition with respect to conformal-time translations), not via a prior type-III_1 identification. The Bianchi IX situation is harder: the anisotropic β-dynamics breaks the FLRW conformal-time symmetry, so the KLS shortcut is also unavailable.

Three partial strategies toward TBD-1. Strategy (i): Haag–Kastler axioms + SO(3) homogeneity. If the net $O \mapsto \mathcal{A}(O)$ satisfies the Haag–Kastler axioms (isotony, locality, covariance under the residual SO(3) isometry) and the Bianchi IX quantum field theory satisfies the *Buchholz–Junglas split property* for the homogeneous spatial foliation (i.e., for any two causally complete regions $O_1 \Subset O_2$, there exists an intermediate type-I factor), then standard AQFT theorems [**?**, **?**] imply that the local algebras are type-III factors. The additional step from type-III to type-III_1 requires showing that the Connes S-invariant $S(\mathcal{A}(W_\gamma)) = [0, \infty)$; we argue this below via the BKL ergodicity.

[TBD-1.1: prove] The split property for free scalar fields on Bianchi IX has not been established. The proof would require uniform energy bounds on the quantum field relative to the BIX homogeneous foliation — a non-trivial task given the time-dependent metric (**??**) and the absence of a static time.

Strategy (ii): Connes S-invariant via BKL ergodicity. For any factor $\mathcal{N}$ with faithful normal weight ϕ, the Connes S-invariant is defined as

$$S(\mathcal{N}) \;=\; \bigcap_{\phi} \mathrm{Sp}(\Delta_\phi), \tag{7}$$

where Δ_ϕ is the modular operator of ϕ and the intersection runs over all faithful normal semifinite weights. $\mathcal{N}$ is type-III_1 if and only if $S(\mathcal{N}) = [0, \infty)$, type-$\mathrm{III}_\lambda$ (for $\lambda \in (0, 1)$) if $S(\mathcal{N}) = \{\lambda^n : n \in \mathbb{Z}\}$, and type-$\mathrm{III}_0$ if $S(\mathcal{N}) = \{0, 1\}$.

For the reference state ω_{BIX}, the modular operator $\Delta_{\omega_{\mathrm{BIX}}}$ encodes the "temperature" seen by the observer along γ. The Kasner-bounce sequence in the BKL phase is encoded by the Gauss-shift σ_G with Kolmogorov–Sinai entropy $\lambda_{\mathrm{BKL}} = \pi^2/(6 \log 2) > 0$ (Definition 1). The positive KS entropy means the Gauss-shift is *strongly mixing* and

Bernoulli (Adler–Weiss theorem), so the associated time-evolution has *absolutely continuous spectrum* on $L^2([0,1], \mu_{\text{Gauss}})$. If this continuous spectrum is reflected in the modular spectrum $\text{Sp}(\Delta_{\omega_{\text{BIX}}})$ via the Manin–Marcolli geodesic-flow representation [9], one expects $S(\mathcal{A}(W_\gamma)) = [0, \infty)$, i.e., type-$\text{III}_1$.
[**TBD-1.2: prove**] The link between BKL Gauss-shift KS entropy (a classical dynamical systems result) and the quantum modular spectrum $\text{Sp}(\Delta_{\omega_{\text{BIX}}})$ is conjectural. Making it precise requires:

(a) Identifying ω_{BIX} as a KMS state (or quasi-free state) for some BIX-adapted dynamics;

(b) Showing the modular operator $\Delta_{\omega_{\text{BIX}}}$ has continuous spectrum via the Manin–Marcolli embedding $\mathcal{A}(W_\gamma) \hookrightarrow B(L^2(X(2), \mu_{\text{Gauss}}))$ (which is itself Conjecture M45.1.C);

(c) Ruling out III_λ ($\lambda \in (0,1)$) — i.e., that the spectrum is the full $[0, \infty)$ and not a discrete subgroup $\{\lambda^n\}$.

For step (c), the aperiodicity of the Gauss shift (its spectrum is purely continuous on L^2 with no point masses at $\{\lambda^n\}$) is encouraging but does not automatically imply III_1 for the quantum algebra.

Strategy (iii): Mixmaster scrambling as surrogate for geometric modular flow. In de Sitter, the de Sitter temperature $T_{dS} = H/2\pi$ controls the modular flow rate, with T_{dS} arising from the Killing horizon structure. For Bianchi IX, there is no Killing horizon, but the Mixmaster oscillations produce an effective "scrambling temperature" at rate λ_{BKL}. One may *conjecture* (heuristically) that the modular flow of $\mathcal{A}(W_\gamma)$ in ω_{BIX} is *ergodic relative to the Gauss-measure foliation of* $X(2)$, and that this ergodicity — in the quantum context — forces $S(\mathcal{A}(W_\gamma)) = [0, \infty)$.
[**TBD-1.3: prove**] This heuristic requires: identifying a *geometric* realisation of the modular Hamiltonian K_{BIX} as a differential operator on $L^2(X(2), \mu_{\text{Gauss}})$; and showing K_{BIX} has purely continuous spectrum. A possible route is the Bost–Connes arithmetic algebra associated with $\Gamma(2)$ (Connes–Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*, Ch. III), where the modular Hamiltonian is related to the geodesic flow generator on $X(2)$ and the spectrum is known to be continuous. This route depends on Conjecture M45.1.C and is therefore circular without an independent proof of Step C.

Summary of TBD-1 sub-markers

Marker	Dependency	Status
TBD-1.1	Split property for free scalar on BIX homogeneous foliation	Open (AQFT/curved spacetime)
TBD-1.2	Gauss-shift continuous spectrum $\Rightarrow S(\mathcal{A}) = [0, \infty)$	Open (requires steps (a)–(c) abov
TBD-1.3	BIX modular Hamiltonian = geodesic flow generator on $X(2)$	Open (circular with Conj. M45.1.

The most tractable near-term route is Strategy (i) + partial (ii): establish the Haag–Kastler net and split property on BIX homogeneous slices (using SO(3) covariance and energy bounds for the free scalar in a time-dependent homogeneous background), obtain the type-III label, and then use the Gauss-shift ergodicity as heuristic input to argue III_1 rather than III_λ. A rigorous proof of type-III_1 for any non-Killing-symmetric cosmological background would be a significant result in AQFT on curved spacetime in its own right, independent of the applications to crossed products pursued here.

4.2 Step B: Crossed product

Given Conjecture 2, Tomita–Takesaki provides the modular automorphism $\sigma_t^{\phi_\gamma}$ for any KMS state ϕ_γ, e.g., the Hartle–Hawking-analogue state on M_{BIX} (whose existence is itself non-trivial in Bianchi IX; see [13]).

Conjecture 3 (Speranza transfer, M45.1.B)**.** *The Speranza intrinsic-observer crossed-product construction* (4) *applies to* M_{BIX} *with* τ*-foliation, yielding a type-*II_∞ *factor* $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty} = \mathcal{A}_{\mathrm{obs}} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}$ *with semifinite trace* τ_{II}*.*

[**TBD-2: prove**] Speranza's flow-of-weights argument uses cosmological time. Bianchi IX has no global cosmological time: only the Misner volume $\tau = -\frac{1}{3}\log V$ plays a comparable role, and τ is monotonic only in the BKL backward direction. The argument must handle the non-globality of τ.

4.3 Step C: Modular-flow = Manin–Marcolli geodesic flow

This is the load-bearing identification.

Conjecture 4 (Flow identification, M45.1.C)**.** *On* $\mathcal{H}_{\mathrm{BIX}} \cong L^2(X(2), \mu_{\mathrm{Gauss}}) \otimes \ell^2$*, the modular flow* $\sigma_t^{\phi_\gamma}$ *of Conjecture 3 acts (up to inner automorphism on the* ℓ^2 *factor) as*

$$\sigma_t^{\phi_\gamma}(f) \;=\; f \circ g_{-ct}, \qquad f \in L^2(X(2), \mu_{\mathrm{Gauss}}), \tag{8}$$

where g_t *is the Manin–Marcolli geodesic flow on* $X(2)$ *and* $c > 0$ *is a normalisation constant identified below.*

[**TBD-3: prove**] The identification of the Speranza-Connes-Takesaki flow with the Manin–Marcolli geodesic flow is the deepest open step. A possible route: realise $\mathcal{A}_{\mathrm{obs}}$ as a sub-algebra of the Bost–Connes type arithmetic algebra associated with $\Gamma(2)$ (Connes–Marcolli *Noncommutative Geometry, Quantum Fields and Motives*, Ch. III), where the modular flow has an intrinsic geometric realisation by the geodesic flow on $X(2)$.

4.4 Step D: Modular-flow rate

Given Conjecture 4, the rate of $\sigma_t^{\phi_\gamma}$ relative to the natural Gauss-measure parametrisation is $\lambda_{\mathrm{mod}} = 2\pi/\beta_{\mathrm{eff}}$ where β_{eff} is the inverse Hawking-like temperature of the Bianchi IX intrinsic observer.

Conjecture 5 (Rate identification, M45.1.D)**.** *The normalisation constant* c *in* (5) *is fixed by*

$$\lambda_{\mathrm{mod}} \;=\; 2\pi\,\lambda_{\mathrm{BKL}} \;=\; \frac{\pi^3}{3\log 2} \;\approx\; 14.907 \quad \textit{(natural BKL units)}, \tag{9}$$

where $\lambda_{\mathrm{BKL}} = h_{\mathrm{KS}}(\sigma_{\mathrm{Gauss}}) = \pi^2/(6\log 2)$ *from Definition 1.*

[**TBD-4: prove**] The factor 2π comes from the Bisognano–Wichmann normalisation $\beta_{\mathrm{BW}} = 2\pi$ in the absence of explicit boost generator.

5 Modular Krylov complexity bound

Combining Conjectures M45.1.A–M45.1.D with the modular-Krylov formalism of Caputa–Magán–Patramanis–Tonni [3] and the universal Lanczos-cosh bound of Parker–Cao–Avdoshkin–Scaffidi–Altman [2] yields a falsifiable target.

Conjecture 6 (Bianchi IX modular Krylov bound)**.** *For any operator $\mathcal{O} \in \mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ with finite modular two-point function $(\mathcal{O}, \mathcal{O})_{\phi_\gamma} < \infty$, the modular Krylov complexity satisfies*

$$\mathcal{C}_K^{\mathrm{mod}}(\mathcal{O}; s) \;\leq\; A_{\mathcal{O}} \;\cosh\!\big(2\pi\, \lambda_{\mathrm{BKL}}\, s\big), \tag{10}$$

with $A_{\mathcal{O}}$ a state-dependent prefactor and $\lambda_{\mathrm{BKL}} = \pi^2/(6 \log 2)$. The corresponding modular Lyapunov exponent is bounded by $\lambda_K^{\mathrm{mod}} \leq 2\pi\, \lambda_{\mathrm{BKL}}$.

[**TBD-5: prove**] The Parker–Cao argument in [2] requires linearly growing Lanczos coefficients $b_n \sim \alpha n$. Their argument transposes from Hamiltonian to modular evolution (this is the Caputa et al. [3] step) but the *slope* $\alpha = \pi \lambda_{\mathrm{BKL}}$ is conjectural.

Counter-example caveat. Bhattacharjee–Cao–Nandy–Pathak [14] exhibit saddle-dominated scrambling systems where Krylov complexity grows exponentially without underlying chaos. In Bianchi IX, the BKL stochasticity is genuine (Gauss-shift entropy is positive); but the *converse* statement — saturation of (7) implies BKL chaos — is *not* claimed here.

Effective temperature. Define the effective inverse temperature of the Bianchi IX intrinsic observer by

$$\beta_{\mathrm{eff}} \;:=\; \frac{2\pi}{\lambda_{\mathrm{BKL}}} \;=\; \frac{12 \log 2}{\pi} \;\approx\; 2.6479. \tag{11}$$

This is a *prediction*: the intrinsic Hawking-like temperature of an observer in the BKL phase of Bianchi IX is $T_{\mathrm{eff}} = 1/\beta_{\mathrm{eff}} \approx 0.3777$ in natural BKL units. The von Neumann coupling constant of $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ (in the Murray–von Neumann sense) is conjecturally proportional to $\lambda_{\mathrm{BKL}}^{-1}$.
[**TBD-6: prove**] The proportionality constant in the von Neumann coupling identification is undetermined and constitutes the main content of the rate conjecture beyond the symbolic-dynamics input.

6 Connection to Lemma A.1 (paper #5)

In [15] we proved (Lemma A.1) that the set of configurations in Bianchi IX vacuum with eigenvalue crossings of the spatial Ricci tensor has Lebesgue measure zero in the Misner configuration space. This is consistent with Conjecture M45.1: eigenvalue crossings correspond to non-generic Kasner-bounce orbits, which form a measure-zero subset of $X(2)$ under μ_{Gauss} (the periodic orbits of σ, dense but countable).

Compatibility check. The Fan–Fathizadeh–Marcolli [10] expansion of the Connes spectral action on Bianchi IX in modular forms of weight $2, 14, 18$ should restrict on $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ to a graded trace-class component; this is a non-trivial consistency check on Conjecture M45.1.

7 Falsifier protocol

The construction above is falsifiable. A concrete numerical protocol (developed in detail in the companion note `falsifier_protocol.md`) consists of:

(F1) *Numerical verification of* λ_{BKL}. Iterate the Gauss shift $\sigma : [0,1] \to [0,1]$, $\sigma(x) = \{1/x\}$, on $N = 10^6$ trajectories of length $T = 10^4$, sample the Lyapunov rate via finite-time Birkhoff averages, and verify $\lambda_{\mathrm{BKL}} = \pi^2/(6\log 2) \approx 2.3731$ to four significant figures. (This is a textbook exercise; serves as sanity check.)

(F2) *Discretised Bianchi IX Krylov rate.* Take a truncation of the WDW (Wheeler–DeWitt) Hamiltonian on Bianchi IX in Misner variables to $n_{\mathrm{rank}} = 10^3$ basis modes, compute the Lanczos coefficients b_n of a generic local field operator under the truncated modular evolution, and extract the Krylov-complexity growth rate $\lambda_K^{\mathrm{trunc}}$.

(F3) *Rate comparison.* Conjecture 6 predicts $\lambda_K^{\mathrm{trunc}} \leq 2\pi\,\lambda_{\mathrm{BKL}} \approx 14.907$, with saturation in the BKL-chaotic regime and violation only in the (measure-zero) periodic-Kasner orbits. A numerical Lyapunov rate exceeding 14.907 by more than the truncation error would falsify Conjecture M45.1.

(F4) *Lemma A.1 compatibility.* Verify that the periodic-orbit subset of $X(2)$ (of σ-measure zero) maps to the eigenvalue-crossing set of paper #5 Lemma A.1 under the Misner-to-modular-surface change of variables.

8 Outlook

- *Lean formalization.* Step A (§4.1) and the Gauss-shift entropy computation (3) are tractable in mathlib (continued fractions and ergodic theory both exist). Steps B, C, D require von Neumann algebras (sparse in mathlib as of 2026).
- *Extension to Bianchi class A (types VIII, VI$_{-1/9}$).* The Manin–Marcolli framework extends [9] to all class A Bianchi types; the construction here should also extend.
- *Genus-$g \geq 2$.* The companion paper #4 (Modular Shadow LMP v2.5) noted that the genus-$g \geq 2$ extension fails for the FLRW case. The Bianchi IX setting is intrinsically tied to $X(2)$ (genus zero) and so the question is moot here; but extension to higher level $X(N)$ is interesting.
- *Co-authors targeted.* Matilde Marcolli (Caltech, for the symbolic-dynamics half) and Antony Speranza (Maryland, for the intrinsic-observer construction); see the companion `outreach_strategy.md`.

Discipline statement

- All 14 arXiv references in the bibliography below were WebFetch-verified on 2026-05-06 against `arxiv.org` directly.
- Six [**TBD: prove**] markers are flagged in §4–5; this paper proves *none* of the four conjectures M45.1.A–M45.1.D.

- No large-language-model output was treated as authoritative without independent live re-verification (Mistral strict-banned per project policy).
- This is a *conjecture paper*: the Manin–Marcolli correspondence (1) is a theorem, the Speranza/Witten/CLPW crossed-product framework is established, but the identification $\sigma_t^{\phi_\gamma} \leftrightarrow g_t$ (Conjecture 4) is the load-bearing open step.

References

[1] J. Maldacena, S. H. Shenker, D. Stanford, *A bound on chaos*, JHEP **08** (2016) 106, arXiv:1503.01409.

[2] D. E. Parker, X. Cao, A. Avdoshkin, T. Scaffidi, E. Altman, *A Universal Operator Growth Hypothesis*, Phys. Rev. X **9**, 041017 (2019), arXiv:1812.08657.

[3] P. Caputa, J. M. Magán, D. Patramanis, E. Tonni, *Krylov complexity of modular Hamiltonian evolution*, Phys. Rev. D **109**, 086013 (2024), arXiv:2306.14732.

[4] E. Witten, *Notes on Some Entanglement Properties of Quantum Field Theory*, Rev. Mod. Phys. **90**, 045003 (2018), arXiv:1803.04993.

[5] E. Witten, *Gravity and the Crossed Product*, JHEP **10** (2022) 008, arXiv:2112.12828.

[6] V. Chandrasekaran, R. Longo, G. Penington, E. Witten, *An Algebra of Observables for de Sitter Space*, JHEP **02** (2023) 082, arXiv:2206.10780.

[7] J. Kudler-Flam, S. Leutheusser, G. Satishchandran, *Algebraic Observational Cosmology*, arXiv:2406.01669 (2024).

[8] A. J. Speranza, *An intrinsic cosmological observer*, arXiv:2504.07630 (2025).

[9] Yu. Manin, M. Marcolli, *Symbolic Dynamics, Modular Curves, and Bianchi IX Cosmologies*, Ann. Fac. Sci. Toulouse Math. **25** (2016), arXiv:1504.04005.

[10] W. Fan, F. Fathizadeh, M. Marcolli, *Modular forms in the spectral action of Bianchi IX gravitational instantons*, JHEP **04** (2020) 087, arXiv:1511.05321.

[11] T. Damour, M. Henneaux, H. Nicolai, *Cosmological Billiards*, Class. Quant. Grav. **20** (2003) R145, arXiv:hep-th/0212256.

[12] N. Cornish, J. Levin, *The Mixmaster Universe: A Chaotic Farey Tale*, Phys. Rev. D **55** (1997) 7489, arXiv:gr-qc/9612066.

[13] J. M. Heinzle, C. Uggla, *A new proof of the Bianchi type IX attractor theorem*, Class. Quant. Grav. **26** (2009) 075016, arXiv:0901.0806.

[14] B. Bhattacharjee, X. Cao, P. Nandy, T. Pathak, *Krylov complexity in saddle-dominated scrambling*, JHEP **05** (2022) 174, arXiv:2203.03534.

[15] K. Remondière, *Eigenvalue-crossing measure-zero lemma for Bianchi IX vacuum configurations* (paper #5, in preparation, 2026).

[16] J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field*, J. Math. Phys. **16** (1975) 985; *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976) 303.

[17] D. Buchholz, E. H. Wichmann, *Causal independence and the energy-level density of states in local quantum field theory*, Commun. Math. Phys. **106** (1986) 321–344.

[18] D. Buchholz, R. Verch, *Scaling algebras and renormalization group in algebraic quantum field theory*, Rev. Math. Phys. **7** (1995) 1195–1240.

[19] M. De Clerck, S. A. Hartnoll, and J. E. Santos, *Mixmaster chaos in an AdS black hole interior*, JHEP **07** (2024) 202, arXiv:2312.11622.